

4 - Week FACULTY QUANTUM IMMERSION

Hands-On Experimental Quantum Training | June 2026



Inside the Immersion:

- Build a Magneto Optical Trap (MOT)
- Run optical and quantum experiments
- Complete 28 hours of high & ultra-high vacuum training
- Train inside a workforce-aligned quantum lab*

Limited to 6 Faculty

Selected by program faculty

- ✓ Tuition covered
- ✓ Institutions cover travel & lodging
- ✓ Includes 2-Day Educators Workshop

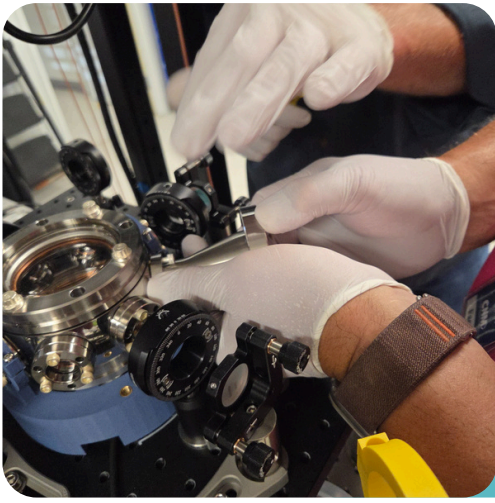
Apply Now

PRE-REGISTRATION REQUIRED



Quantum Learning Lab
Exclusively Offered in Albuquerque

*See attached equipment list



Quantum Learning Lab Equipment Overview

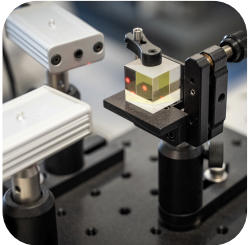


Magneto-Optical Traps

- Four Rb-85 MOTs build from the Vescent Laser and the UHV chambers

Lasers

- Vescent D2 780nm (108mW) Laser and Control Electronics
- Labs Electronics DInsec pulsed 520nm laser (down to 300ps)
- ThorLabs NPL45B 450nm diode pulsed laser (5-39ns)
- ThorLabs NPL52C 520nm diode pulsed laser (6-129ns)
- ThorLabs L405P20 405 nm, 20mW laser diode
- ThorLabs L658P040 658nm, 40mW laser diode
- Compact Laser Modules at 405nm, 532nm, 635nm, and 639nm



Optics

Four optical tables

Full suite of optical components (lens, mirrors, gratings, opto-mechanical mounts)

Polarized and non-polarized beam splitter (visible and NIR)

780nm EOM (3 GHz) and AOM (80 MHz and 200 MHz)

BBO Crystals (Type I and Type II) 405nm to 810 nm



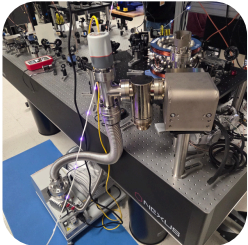
ThorLabs Education Kits

Michelson Interferometer

Portable Optical Tweezers

Time Resolved Absorption Spectroscopy

Quantum Optics with Polarized Entanglement Expansion



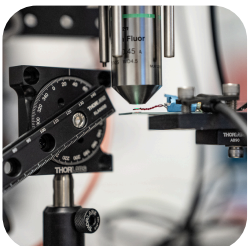
Vacuum System

Three high vacuum systems (10^{-7} torr) with 1 cubic foot chambers

One ultra-high vacuum 6-inch spherical-octagonal chamber (10^{-10} torr) with Ion / TSP pump

Three ultra-high vacuum 4.5-inch spherical-octagonal chambers (10^{-9} torr) with Ion Pumps.

Helium Leak Checking System



Sandia National Labs - Qick-Dawg diamond nv microscopy/magnetometer

Binder 14.1 cubic ft (39"x31"x20") backout oven. Range: ambient to 300 degrees C. Temp

Fluctuation of 0.7 C @ 150 C